

Austin Robinson

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EDUCATION

University of North Carolina at Chapel Hill

May 2026

B.S. Computer Science

CHAPEL HILL, NC

- **Relevant Coursework:** Data Structures, Algorithms & Analysis, Intro to Probability & Statistics, Data Science & Econometrics, Artificial Intelligence, System Security, and Foundations of Software Engineering

TECHNICAL SKILLS

- **Programming Languages (ordered by proficiency):** Python, Java, C++, Typescript/JavaScript, C, R
- **Frameworks/Tools:** React.js, Angular, Node.js, Flask, **Docker**, Kubernetes, AWS (EC2), scalable application architecture, **RESTful API design**, Maven, **Spring Boot**
- **AI-Assisted Development:** Use AI Coding/productivity tools to accelerate debugging, prototyping, and workflow throughout development.
- **Data & Analysis:** SQL, PostgreSQL, SQLAlchemy, Pandas, NumPy, scikit-learn, JSON, Matplotlib

EXPERIENCE

Vechter Home Solutions

2026 – Present

Application Developer

Remote

- Built and shipped a full-stack field diagnostics application enabling sales reps to run on-site tests on homeowners' solar and backup power systems and receive automated system recommendations, in active daily field use.
- Delivered a custom, end-user-facing solution supporting both internal sales staff and field technicians, contributing to a **\$100K–\$250K weekly increase** in closed sales revenue.
- **Owned the application end-to-end.** Architecture, deployment, and ongoing support, for a non-technical user base, including troubleshooting and iteration based on direct field feedback.

ACADEMIC/PERSONAL PROJECTS

LearnWithAI – AI-Powered Lecture Summarizer | Angular, FastAPI, RabbitMQ, Kubernetes

2026

- **Collaborated in a 4-person team** to engineer a modular learning management platform enabling students to upload lecture audio/text and receive automated AI summaries via **Azure OpenAI**.
- **Implemented a real-time event system via WebSockets**, streaming background worker job statuses directly to the frontend client with an automated HTTP polling fallback mechanism.
- **Architected a type-safe backend infrastructure using SQLAlchemy** for data persistence in PostgreSQL, securing endpoints.
- Containerized the application ecosystem using **Docker** and established non-mutating validation pipelines (**Ruff, Pyright, Prettier**), deploying the final infrastructure onto an **OKD Kubernetes** cluster.

Formal Verification of Client-Side Anti-Cheat System | C++, Alloy, SAT Solvers

February – May 2026

- Conducted a formal security analysis of the *Ultimate AntiCheat* open-source framework by **abstracting C++ source code** into a relational state-transition model.
- Defined critical security properties using First-Order Logic and LTL to **verify system integrity against memory** modification, debugger attachment, and unauthorized module injection.
- Utilized the **Alloy Analyzer** to perform automated **SAT-based model checking**, exhaustively exploring state spaces to identify logic flaws and potential memory "leakage" that could bypass detection.
- **Identified and documented counterexamples** where legitimate system states could be spoofed or where detection mechanisms failed to trigger enforcement, providing **mathematically grounded security assurances**.

Extro – Stock Market Forecaster | Next.js, TS, Graphs

June 2025 - Present

- Developed a full-stack financial dashboard where users log in and simulate long-term portfolio growth using fictional balances.
- Integrated **OpenAI API** using **Flask** in the backend to generate personalized investment advice. The AI analyzes user-selected companies, provides performance feedback, profit summaries, and recommendations on whether to invest.
- Implemented secure authentication and data persistence using **Clerk (login)** and a **Flask + SQL RESTful API architecture deployed on AWS EC2**.
- **Result:** Built an intelligent, cloud-ready simulation platform combining AI reasoning with financial analytics to enhance user engagement and learning in investing.